

Attributional Semantics and Hypergraph Modeling in English Translations of Homer’s *Odyssey*

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Abstract

This methods and data paper presents a reproducible framework for tracking attributional semantics in English translations of Homer. It replaces arbitrary passage bins with native Book I–XXIV anchors, extracts more than 13,000 auditable attribution events from six public-domain translations (1616–1919), and models their target–attribute structures as hypergraphs with edge-dependent vertex weights. The release demonstrates how canonical targets, normalized attributes, semantic categories, and source-aligned book anchors can be integrated in a single computational pipeline. Its chronological test finds no robust evidence for monotonic semantic-category drift in this six-text corpus after multiple-testing adjustment.

1 Introduction

The tracking of Homeric epithets across English translations presents a formidable challenge for computational literary studies. Keyword matching and string-distance metrics do not adequately capture the complex, variable syntax of poetic translations. A translator might render a single Greek compound adjective as a multi-word relative clause, an appositive phrase, or omit it entirely. Consequently, the search for “adjective-noun pairs” is an inadequate proxy for the true semantic landscape of the translated *Odyssey*.

This paper serves primarily as a methods and data description rather than a literary interpretation. It details a comprehensive computational protocol applied to a public-domain corpus of six English translations. To address the limitations of keyword matching, we advocate for a shift toward explicitly structured semantic events. Rather than searching for specific lexical strings, the implemented release extracts a deterministic relationship between a canonical target and its assigned attribute. We also discard arbitrary, fixed-length passage grids in favor of native Book I–XXIV source anchors, ensuring the unit of analysis respects the poem’s inherent structure. Finally, we model these translations as hypergraphs with edge-dependent vertex weights, a topological approach suited to capturing the overlapping allocation of attributes to Homeric targets.

2 Corpus and Native-Book Alignment

The foundation of this study is a strictly public-domain corpus comprising six distinct English translations of the *Odyssey*, spanning three centuries: George Chapman (1616), Alexander Pope (1726), William Cowper (1791), S. H. Butcher and Andrew Lang (1879), Samuel Butler (1900), and A. T. Murray (1919). This selection provides a robust basis for engineering, historical-language validation, and testing the resilience of the extraction pipeline across both verse and prose forms.

The six texts contain 735,316 retained English words after poem-only preparation. The final release records 144 translation–book units (six translations times 24 books) and 13,226 extracted

attribution events. Table 1 reports the size and unweighted event density of each translation. The contrast in density is analytically useful but cannot be read as a direct measure of literary richness: it partly reflects verse and prose form, lexical compression, the curated ontology, and the transparency-oriented extractor itself.

Table 1: Verified corpus scope and attribution-event density in the final public-domain release.

Translation	Year	Form	Words	Events / 10,000 words
Chapman	1616	verse	133,440	163.0
Pope	1726	verse	108,596	212.3
Cowper	1791	verse	109,201	172.2
Butcher & Lang	1879	prose	134,673	207.1
Butler	1900	prose	117,320	129.8
Murray	1919	prose	132,086	193.4

2.1 Poem-Only Extraction

A significant obstacle in comparative translation analysis is the presence of editorial paratext. Printed editions and digitized texts (such as those from Project Gutenberg) are replete with introductions, footnotes, marginalia, and publisher advertisements. If left unaddressed, this material severely skews lexical and semantic frequencies.

To resolve this, we implemented a deterministic, poem-only extraction protocol. For EPUB sources, we utilize the package document’s spine to navigate the text, explicitly filtering out non-poem documents and identifying book boundaries based on audited HTML class selectors. For PDF sources (specifically the Murray 1919 edition), we apply a layout-aware classification algorithm that distinguishes printed verse from lower-margin explanatory notes and Greek source text based on character thresholds and bounding-box coordinates.

2.2 Native Book I–XXIV Anchors

Previous iterations of this research attempted to align translations by forcing them into an artificial grid of 480 equal-length passage bins. While computationally convenient, this approach fractures the narrative and introduces severe alignment artifacts when comparing concise prose against expansive verse.

We have entirely replaced the artificial passage grid with native Book I–XXIV source anchors. Every extracted attribution event is mapped to one of the 24 canonical books. This ensures that pairwise comparisons are structurally sound and philologically meaningful.

The change in granularity is not a cosmetic relabelling. An equal-length bin may begin in the middle of an action sequence in one translation and in the middle of a council scene in another, particularly when verse expands or compresses the source differently from prose. Book-level anchoring does not make translations textually identical, but it places each comparison within a shared narrative architecture: the return to Ithaca, the Phaeacian books, the Cyclops episode, and the recognition scenes remain coherent analytical contexts. It also gives readers an intelligible unit for audit and close reading. A contested result can be returned to a named book, a source page or EPUB document, and a local sentence context rather than to a machine-imposed fraction of a file.

Book anchors are large enough to remain stable across strongly divergent translation lengths and small enough to support resampling. The release therefore samples the 24 anchors jointly across translations in its bootstrap chronology analysis, rather than treating thousands of adjacent sentence fragments as independent observations.

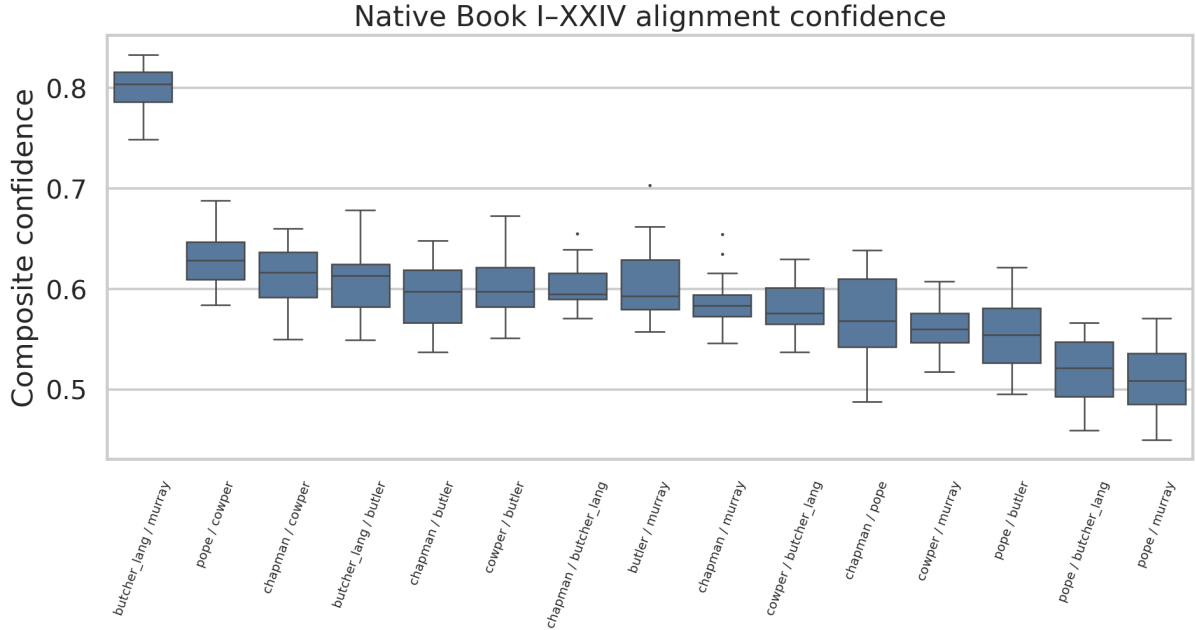


Figure 1: Pairwise alignment confidence across the 24 native books. The stability of the alignment demonstrates the superiority of native-book anchors over arbitrary passage binning.

As demonstrated in Figure 1, the alignment confidence between translation pairs remains remarkably stable when evaluated at the native-book level, providing a reliable foundation for the subsequent semantic extraction.

3 Extraction and Semantic Categorization

The computational core of the implemented release is a deterministic, rule-based attribution extractor. It identifies canonical targets through an audited entity-alias inventory and candidate attributes from a curated semantic ontology. It then assigns the nearest structurally plausible target according to a hierarchy of transparent relations: a nearby prepositive modifier, a copular predicate, a postpositive or local apposition, or a bounded pronoun-carryover heuristic.

Concrete records illustrate how the pipeline moves from text to analytical representation. In Butcher and Lang’s Book III, the sentence “Then wise Telemachus answered him” yields the event TELEMACHUS–“wise,” classified as *cognitive evaluation* and recorded as a high-confidence prepositive modifier. A later phrase in the same book, “the goodly steadfast Odysseus,” yields ODYSSEUS–“steadfast,” classified as *behavioral disposition*, again as a prepositive modifier. These events increment their respective category totals; in the hypergraph, they also contribute incidence weights to the vertices “wise” in the TELEMACHUS hyperedge and “steadfast” in the ODYSSEUS hyperedge. A third example shows why relation type matters: in Butcher and Lang’s Book II, a pronoun-carryover instance links “steadfast” to PENELOPE. It is retained with lower confidence than the direct modifiers, so its contribution to confidence-weighted category and hypergraph counts is correspondingly reduced.

Each retained event records the canonical target and surface form, canonical attribute and surface form, semantic category, relation type, token distance, and extraction confidence. The confidence score is procedural: it begins with a relation-specific base score, adjusts it when more than one local target competes, and multiplies it by the confidence attached to the source-book segmentation. In analyses where events are weighted, a negated event (e.g., “not noble”) receives a reduced negative contribution.

3.1 The Semantic Ontology

Once extracted, attributes are normalized to their lemmas and mapped to a 12-category ontology. This ontology includes categories such as *Divine* (e.g., godlike, immortal), *Physical* (e.g., strong, beautiful), *Social* (e.g., noble, lordly), and *Emotional* (e.g., angry, sorrowful). By aggregating specific lexical items into these broader semantic categories, we can compare the overarching descriptive choices made by different translators while reducing the noise introduced by mere synonym substitution.

The ontology encodes a scholarly decision about which English expressions count as comparable attributions. For that reason, the release preserves both the normalized label and the original surface string. The resulting analysis can answer questions such as whether a translator’s retained attribution events are more frequently classified as social-status or affect-state descriptions than those of another translation. The release also records a valence label where available and retains explicit negation.

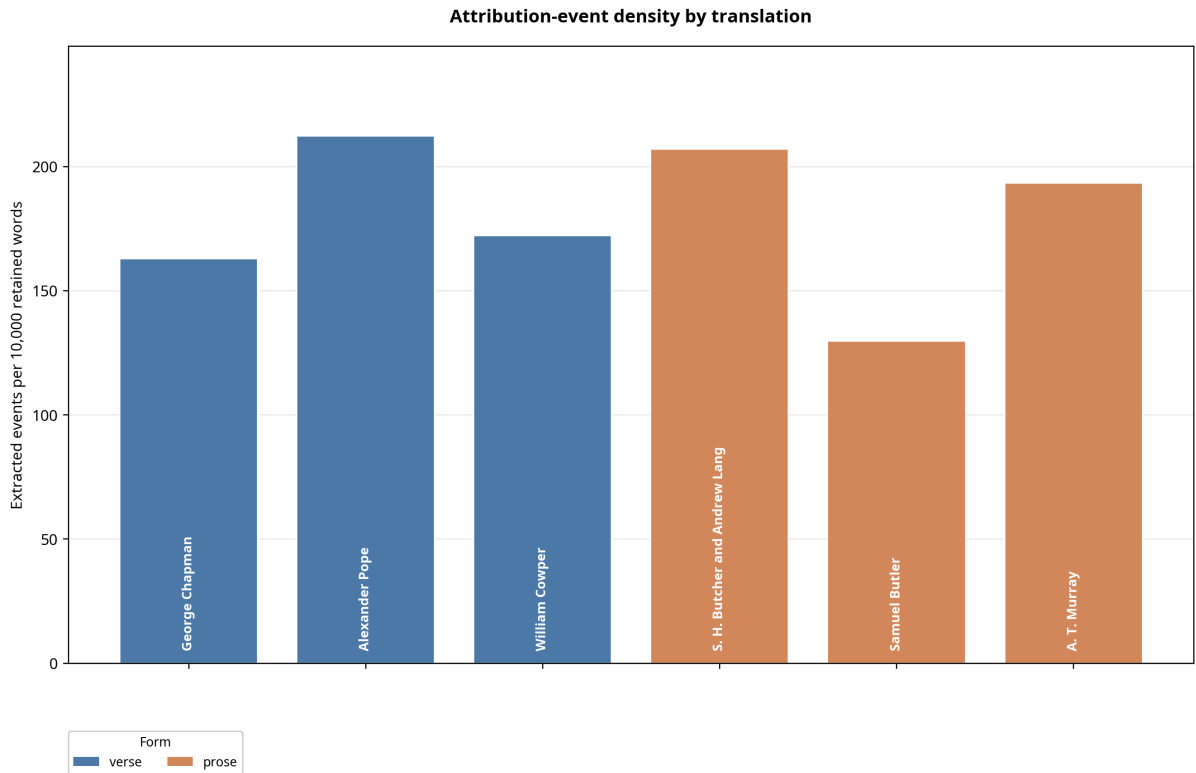


Figure 2: Density of extracted attribution events per 10,000 words across the six translations, distinguished by verse and prose forms.

As shown in Figure 2, the extraction pipeline successfully identifies thousands of attribution events across all six texts. Notably, the overall density of extracted events per 10,000 words varies between translators, reflecting differing approaches to Homeric expansiveness, though the pipeline demonstrates robustness across both verse and prose formats.

Figure 3 displays the distribution of retained events across the 12 semantic categories, making broad commonalities and translator-specific emphases available for comparison. The accompanying event table permits any category pattern to be traced by confidence, relation type, book, target, and original sentence context.

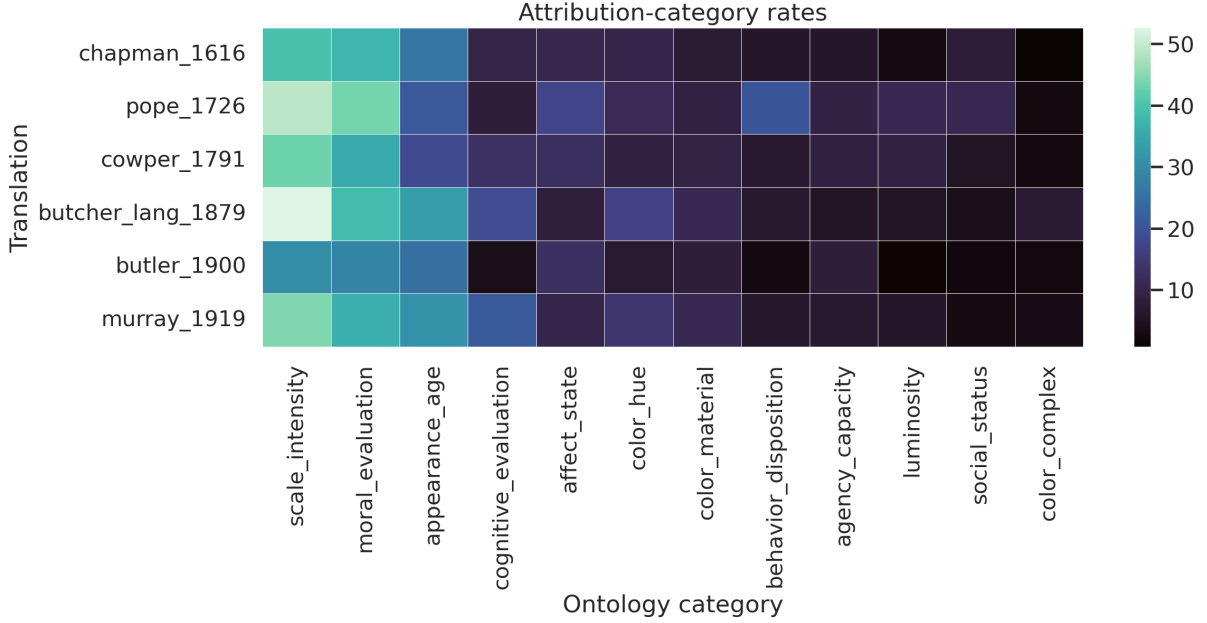


Figure 3: Distribution of extracted attributes across the 12 semantic categories for each translation.

4 Analytical Outputs: Divergence and Chronology

With the attribution events categorized and anchored to native books, the release quantifies semantic distance between translations using pairwise Jensen–Shannon divergence. For a translation t and category c , it forms a confidence-weighted count $w_{t,c}$, normalises it into $p_{t,c} = w_{t,c} / \sum_c w_{t,c}$, and calculates divergence from the midpoint distribution. This produces a compact descriptive measure of differences in category allocation.

The reporting layer distinguishes raw event density from confidence-weighted density. Both are normalized by retained words so that expansive verse and concise prose are not compared as if they supplied the same number of opportunities for local attribution matches. Differences in enjambment, formulaic repetition, and syntactic compression remain substantive features of translation and are retained in the released event records.

Figure 4 shows that the translations form neither a single uniform semantic profile nor a simple chronological sequence. Its pairwise contrasts identify cases for closer inspection alongside the event records and target-level hypergraphs.

4.1 Testing for Chronological Drift

A central question in the study of historical translations is whether semantic choices drift systematically over time. Do later translators consistently prefer certain categories (e.g., *Psychological*) over others (e.g., *Divine*)? We evaluate this hypothesis using exact chronological permutation tests.

Figure 5 shows no robust evidence for monotonic chronological drift in semantic categories across the 1616–1919 window after adjustment. The variance between contemporaneous translators (e.g., Butler and Murray) can exceed the variance observed across centuries, indicating that individual translation strategy is at least as consequential as chronological progression in this corpus.

With only six translations, the chronological test enumerates the possible reorderings of the translation years, reports an exact two-sided permutation probability for each category, and applies Holm adjustment across the 12 categories. The smallest unadjusted result concerns

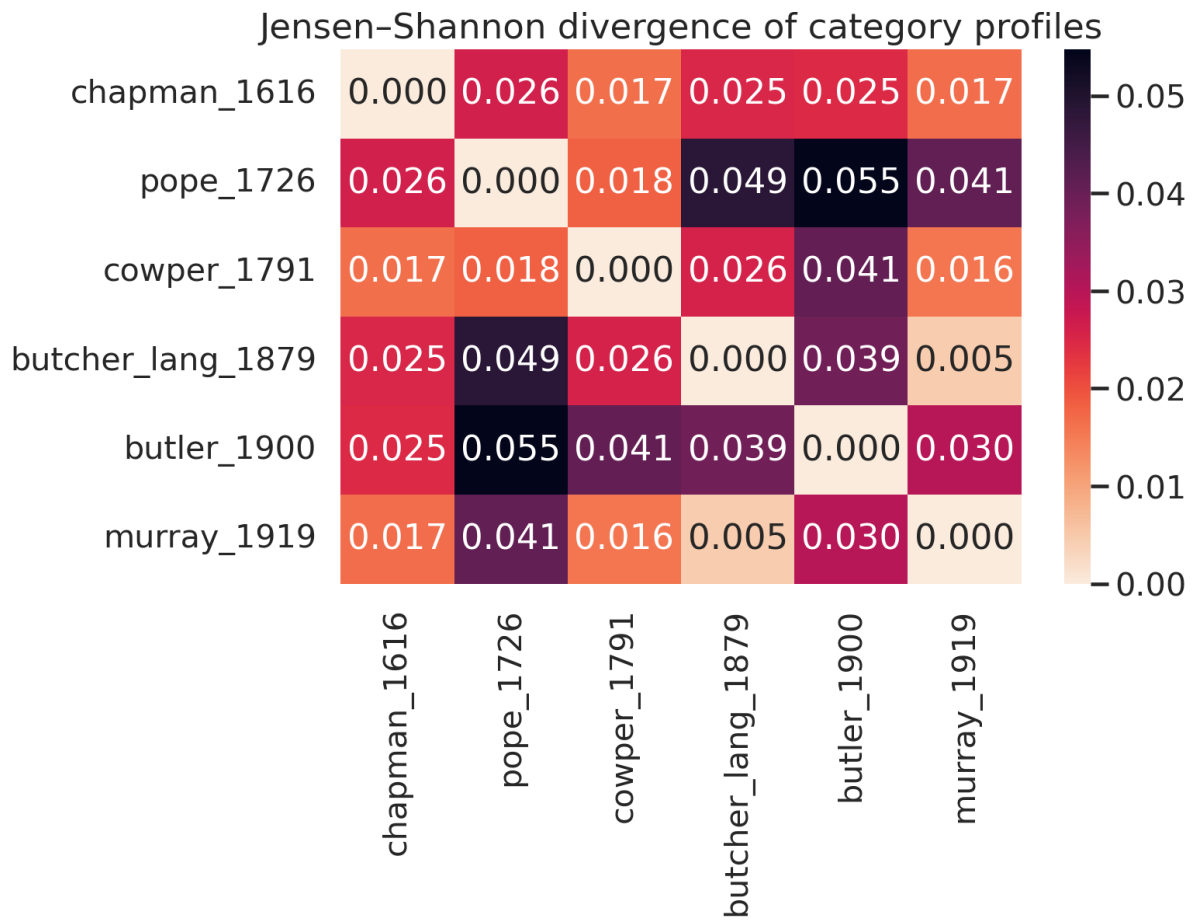


Figure 4: Pairwise Jensen-Shannon divergence of semantic category distributions.

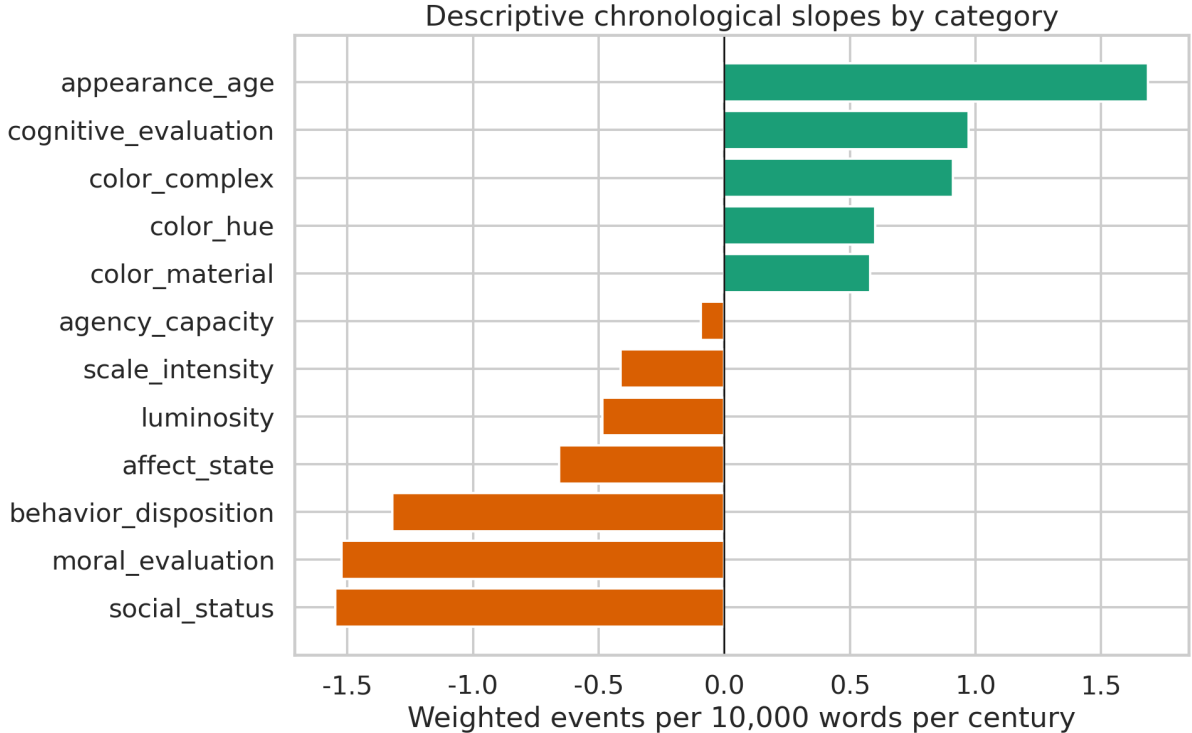


Figure 5: Permutation tests evaluating chronological drift in semantic category usage from 1616 to 1919.

social-status attributions, but it does not remain conventionally significant after adjustment. Consequently, there is no robust evidence for monotonic chronological drift in semantic category usage across this six-text corpus. Book-level bootstrap resampling confirms this finding, demonstrating that apparent directional slopes are often highly sensitive to the narrative distribution of events within a small number of books.

5 Hypergraph Representation of Translations

Traditional network analysis often relies on bipartite graphs to model the relationship between targets and attributes. However, Homeric epithets frequently involve multiple attributes assigned to a single target within a single syntactic event (e.g., “the wise and much-enduring Odysseus”). Bipartite projections force these complex, higher-order relationships into pairwise edges, resulting in a loss of structural information.

To preserve this complexity, we model each translation as a hypergraph with edge-dependent vertex weights. In the present release, a canonical target indexes a hyperedge and normalized attributes are vertices. Crucially, rather than assigning a single scalar weight to an entire hyperedge, we apply edge-dependent vertex weights (EDVWs): the incidence relation between an attribute and its target hyperedge is weighted by the confidence-weighted frequency of that specific attribution event, while the event table preserves the individual contexts from which that aggregate derives. This corresponds to the EDVW formulation in which a vertex may have a distinct weight for each incident hyperedge, reflecting its contribution to that hyperedge [Chitra and Raphael, 2019, Zhu and Segarra, 2023]. This is a practical representation of recurrent attributional structure rather than a claim that a named character literally is a mathematical hyperedge.

The distinction is important. A conventional one-mode projection would connect every pair of attributes that share a target, making “wise” and “much-enduring” appear to have a

direct relation simply because they are both associated with Odysseus. The hypergraph retains the higher-order membership: both terms are linked through an identifiable target and can be examined across individual events and books. Contemporary work on semantic hypergraphs stresses the value of representations that can retain hierarchical and relational richness in natural-language data [Menezes and Roth, 2021]. Likewise, higher-order network research demonstrates that hyperedges can encode group interactions that ordinary dyadic networks discard [Lotito et al., 2022]. The point of importing this language into translation study is not to turn philology into network science, but to avoid discarding the very relational organisation that a translation’s attributions instantiate.

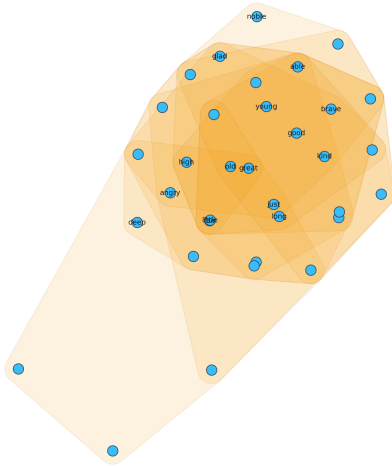
This formal representation is most useful for prioritizing targeted questions: Which targets possess unusually different attribute neighborhoods? Which categories co-occur within a target more in one translation than in another? Which books contribute most to a similarity or divergence?

To ground this abstraction, Figure 6 provides a direct visualisation of the native hypergraph structure for Samuel Butler’s 1900 prose translation. The left panel renders the hyperedge-hull view, in which target entities envelop their associated attribute vertices, explicitly demonstrating how Homeric targets function as higher-order containers for semantic properties. The right panel shows the corresponding bipartite incidence view. For visual clarity, this figure displays only a readability-filtered core of the network (the most frequent targets and their strongest attribute associations) rather than the complete 118-node topology. It serves as an orienting map of the EDVW model’s topological premise rather than an exhaustive semantic census.

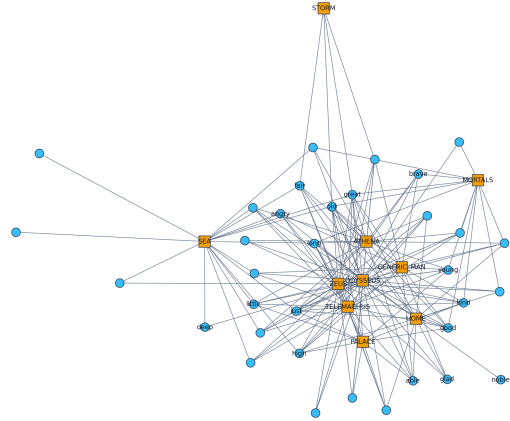
Samuel Butler (1900) — Native target-as-hyperedge visualization

Readable weighted core: 34 of 118 attribute nodes; 10 of 89 target hyperedges; 169 of 821 incidences.

XGI native hyperedge-hull view



XGI bipartite incidence view



Attribute circles and target hyperedges use the colors specified in config/visualizations.json.

Figure 6: A readability-filtered core of the native target-as-hyperedge topology for Samuel Butler’s 1900 translation, visualised using XGI. The hyperedge-hull view (left) and bipartite incidence view (right) illustrate the higher-order assignment of attributes to canonical targets.

This hypergraph model allows for more sophisticated comparative metrics. By analysing incidence matrices and target-centred attribute intersections, we can quantify structural similarity in a way that respects the combinatorial nature of Homeric formulas. Figure 7 supplies a second descriptive lens alongside raw category counts: texts may have similar category totals but allocate those categories to different targets, or conversely retain similar target-level configurations while differing in overall event density. This distinction is valuable precisely because it resists a false choice between purely lexical comparison and a wholly abstract network model.

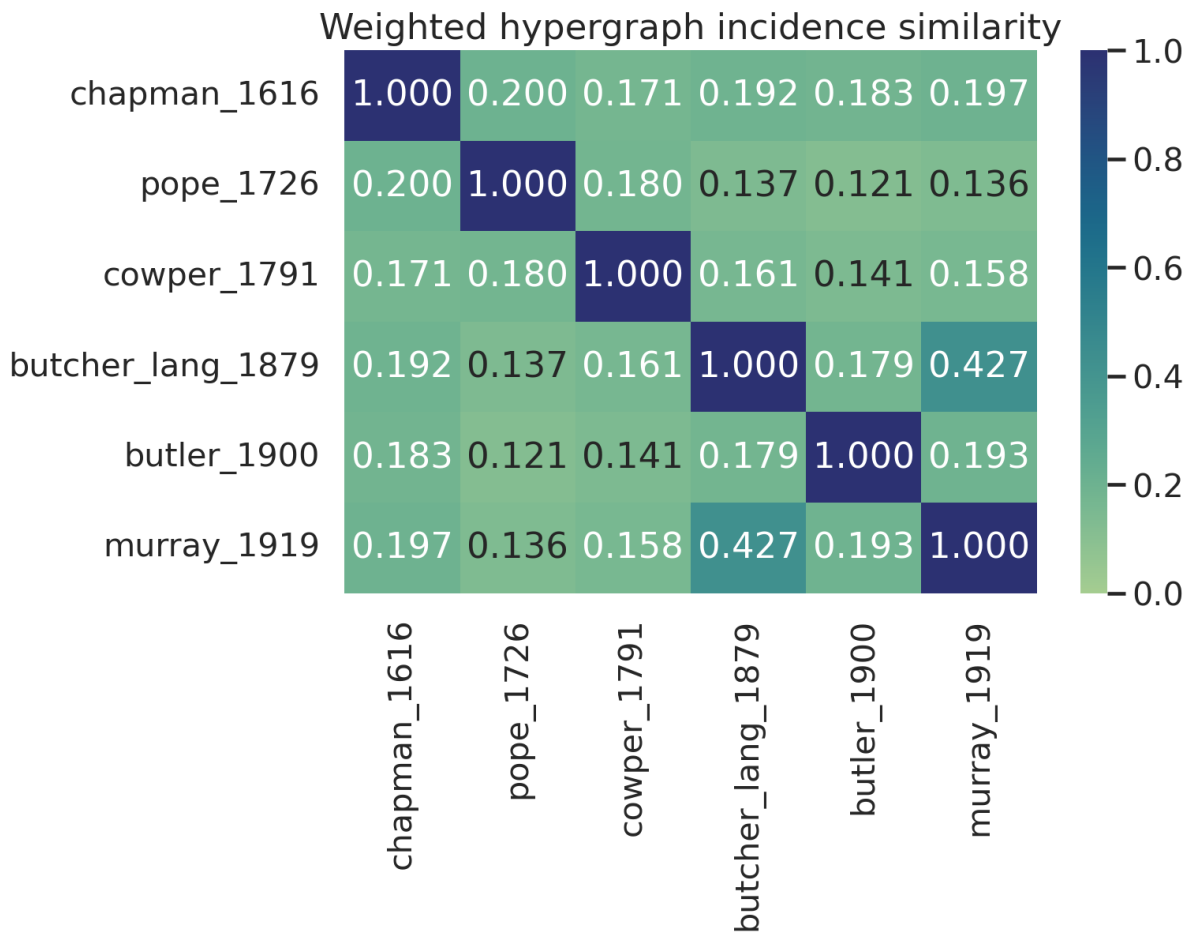


Figure 7: Similarity matrices comparing the hypergraph topologies of the six translations.

6 Target-Level Signatures

Beyond aggregate category distributions, the hypergraph model enables granular analysis of specific canonical targets. How do different translators construct the semantic identity of core entities like Odysseus, Penelope, or Telemachus?

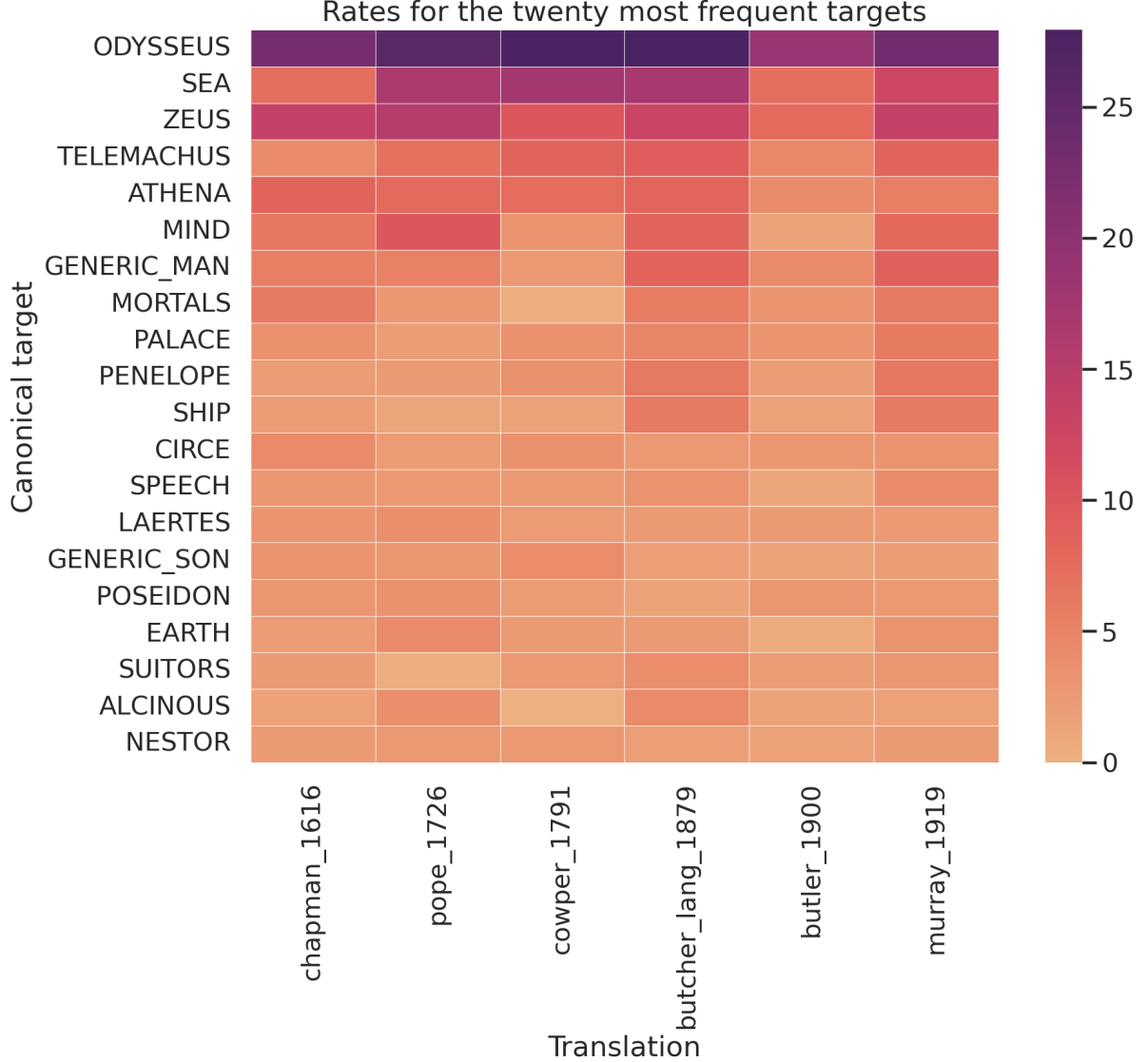


Figure 8: Attribution rates for the most frequently modified canonical targets.

Figure 8 displays attribution rates for the most prominent targets. Native-book alignment makes these target-level signatures comparable across equivalent narrative spans. An aggregate signature can then be decomposed by book and relation type and followed back to the sentence contexts that produced it. For example, a high social-status rate may derive from formulaic titulature, a repeated narrative epithet, or a dialogue-specific phrase; the event records distinguish these mechanisms rather than folding them into a single undifferentiated count.

7 Conclusion

The computational analysis of translation requires methodologies that respect both the philological realities of the text and the structural complexities of the language. By replacing arbitrary

passage grids with audited, poem-only native Book I–XXIV anchors, we have established a structurally sound foundation for comparing English translations of the *Odyssey*. The application of hypergraphs with edge-dependent vertex weights to model attribution events captures the higher-order combinatorial nature of Homeric epithets, offering a significant advance over traditional bipartite projections.

The analysis of the 1616–1919 public-domain corpus demonstrates the robustness of this pipeline across diverse verse and prose styles. While we observe distinct semantic profiles and varying degrees of divergence among the six translators, we find no robust evidence for monotonic chronological drift across this period after adjustment. The variance between contemporaneous translators often exceeds the variance observed across centuries.

This public-domain baseline paves the way for future research. A reproducible public-domain baseline can support comparative work on formulaic language, character construction, narrative pacing, and reception history. It can also function as a control layer for future restricted-text research: modern translations may be evaluated with the same conceptual schema, native-book anchors, and audit practices even where the underlying prose cannot be redistributed. In that sense, the present release offers both a completed methodological demonstration and an extensible research infrastructure.

7.1 Project Basis and Future Corpus Expansion

This paper is based on the openly documented computational project *Odyssey Attributional Semantics*, available at <https://github.com/mboudour/odyssey-attributional-semantics>. In its present form, the paper describes the computational pipeline implemented in that project and reports the findings produced by the current released corpus. Its purpose is therefore methodological and evidentiary: to make explicit how poem-only source texts are prepared, how attributional events are extracted and normalized, how native Book I–XXIV anchors support comparison, and how the resulting tables and hypergraph representations can be read alongside their provenance records. The present analysis should not be understood as an exhaustive account of the English reception of Homer’s *Odyssey*; it is a reproducible account of the corpus and procedures presently available in the project.

Future work is intended to extend this framework through the careful incorporation of additional publicly available English translations for which the legal status and terms of use permit responsible inclusion. Such expansion will require source-specific preparation, verification of poem boundaries and book anchors, and explicit preservation of provenance so that new materials remain comparable without obscuring edition-specific differences. In a later stage, if legally usable transcripts become publicly available, the corpus may also be extended to include the transcripts of Christopher Nolan’s 2026 film *The Odyssey*. Any such addition would require a separately documented preparation and alignment protocol, and it would be analyzed as a distinct adaptation layer rather than treated as interchangeable with a translated poetic text.

References

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